A Highly Clean Recipe Dataset with Ingredient States Annotation for State Probing Task

Taro Yamada

Abstract

Large language models (LLMs) learn from procedural texts but rarely observe real-world phenomena, making it difficult to track implicit state changes in instructions. Existing benchmarks for recipe understanding focus on surface-level tasks such as ordering and alignment, rather than modeling implicit world-state transitions. We introduce NHK Recipe State Probing, a benchmark that directly evaluates and trains models' ability to infer and track ingredient states through cooking procedures. Our dataset features professionally curated recipes from NHK Educational with explicit per-step world-state annotations and ingredient transformation links. We propose three novel tasks: Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing, which assess models' understanding of procedural state changes beyond surface ordering. Comprehensive evaluation across LLMs and large multimodal models (LMMs) reveals that fine-tuning on our annotated data significantly improves large models (e.g., Llama-3.1-70B: 0.558→0.720 average accuracy), while smaller models often degrade. Multimodal models show substantial improvements, with Qwen-2.5-VL-72B achieving 0.919 average accuracy after fine-tuning. Our work provides a high-quality, state-centric benchmark to advance grounded procedural understanding research.

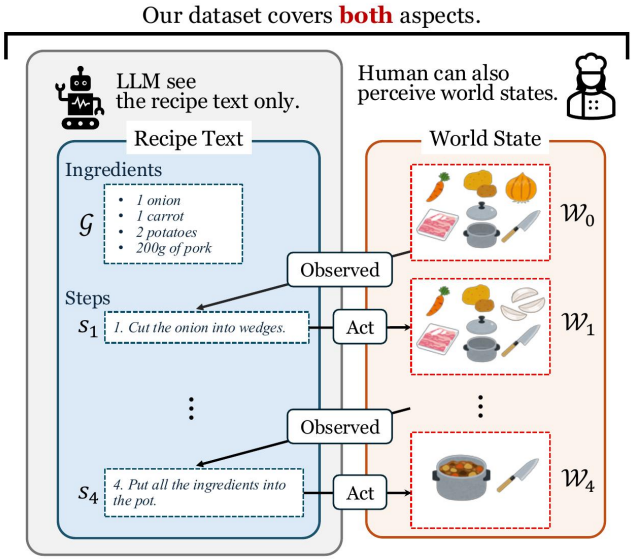

Figure 1: Overview of our approach: We augment high-quality recipe data with explicit state annotations to enable direct evaluation of models' procedural state tracking capabilities.

1 Introduction

Large language models (LLMs) demonstrate remarkable capabilities in understanding and generating procedural text, yet they face significant challenges when required to track implicit state changes throughout multi-step processes. This limitation is particularly evident in cooking recipes, where the transformation of ingredients through various operations creates complex state dependencies that are rarely made explicit in natural language instructions.

Consider a recipe that begins with "Cut the chicken into bite-sized pieces" and later instructs to "Add the chicken to the pan." While humans intuitively understand that the second step refers to the previously cut chicken pieces, current language models struggle to maintain consistent tracking of such ingredient states throughout cooking procedures. This challenge stems from the fact that procedural texts typically omit intermediate states, expecting readers to infer transformations and maintain mental models of evolving ingredient conditions.

Existing benchmarks for recipe understanding, such as RecipeQA [6] and Recipe1M [3], primarily focus on surface-level tasks including step ordering, ingredient alignment, and visual-textual matching. While these tasks provide valuable insights into basic comprehension capabilities, they do not directly assess models' ability to reason about implicit world-state transitions—a fundamental requirement for true procedural understanding.

To address this gap, we introduce **NHK Recipe State Probing**, a novel benchmark that directly evaluates models' capacity to infer and track ingredient states through cooking procedures. Our approach centers on the key insight that effective procedural understanding requires explicit modeling of world-state transitions, where each step transforms the current state into a new configuration of ingredients and intermediate products.

Our benchmark is built upon professionally curated recipes from NHK Educational's "Minna no Kyo no Ryori" collection, featuring approximately 25,000 high-quality Japanese recipes with standardized ingredient names, quantities, and professionally photographed final dishes. From this collection, we carefully annotated 100 recipes with explicit per-step world states and fine-grained ingredient transformation links, creating the first dataset to systematically capture procedural state transitions in cooking.

We propose three complementary tasks that assess different aspects of state tracking: (1) **Step Reference Resolution**, which tests models' ability to resolve implicit references to previous steps; (2) **Ingredient Usage Prediction**, which evaluates understanding of ingredient consumption patterns; and (3) **Ingredient Tracing**, which assesses tracking of ingredient transformations through complex cooking processes.

Our contributions are fourfold: (1) We present the first high-quality recipe dataset with explicit per-step world-state annotations and ingredient transformation links; (2) We introduce three novel tasks that directly probe models' ability to track implicit state

changes in procedural text; (3) We provide comprehensive evaluation across both language-only and multimodal models, demonstrating that fine-tuning significantly improves large models while smaller models often degrade; and (4) We release our dataset and evaluation framework to foster future research in grounded procedural understanding.

2 Related Works

2.1 Recipe Understanding and Multimodal Cooking Datasets

Recipe understanding has emerged as an important application domain for multimodal AI research. Salvador et al. [3] introduced Recipe1M, a large-scale dataset of cooking recipes paired with images, focusing on cross-modal embeddings between recipe text and food images. This work established the foundation for recipe-image retrieval tasks but did not address procedural state tracking.

RecipeQA [6] extended recipe understanding to question answering, proposing tasks such as visual coherence checking, step ordering, and textual cloze completion. While RecipeQA introduced important procedural reasoning elements, its tasks primarily evaluate surface-level comprehension rather than deep state tracking capabilities. The dataset's focus on static image-text alignment and temporal ordering fails to capture the dynamic evolution of ingredient states that our work explicitly addresses.

More recent efforts like CookingQA [**?**] and YouCookII [**?**] have attempted to bridge vision and language in cooking domains, but these datasets emphasize action recognition and video understanding rather than implicit state modeling. CookingQA's procedural questions focus on ingredient identification and cooking method classification, while YouCookII targets temporal action segmentation in cooking videos. Neither dataset provides the fine-grained state annotations necessary for evaluating models' understanding of ingredient transformations and world-state evolution throughout cooking procedures. Our work builds upon these foundations by explicitly modeling ingredient state transitions throughout cooking procedures.

2.2 Procedural Text Understanding

Procedural text understanding has gained significant attention as a challenging domain for language models. Wu et al. [5] proposed multimodal procedural knowledge understanding through instructional manual sequencing, highlighting the importance of temporal reasoning in procedural domains. However, their work focused on ordering and alignment tasks rather than explicit state tracking.

Previous approaches to procedural understanding have primarily emphasized structural aspects such as step ordering, temporal relations, and goal achievement. Our work differs fundamentally by introducing explicit world-state modeling, where each procedural step is annotated with precise input-output state transitions, enabling direct evaluation of models' state tracking capabilities.

2.3 Large Language Models and Multimodal Models

Recent advances in large language models, including the Llama series [2, 4] and Qwen family [7], have demonstrated impressive capabilities in text understanding and generation. The introduction of multimodal variants such as Qwen2.5-VL [1] has enabled joint reasoning over text and visual inputs, making them particularly relevant for recipe understanding tasks that can benefit from final dish images.

Our evaluation encompasses both language-only models and their multimodal counterparts, providing insights into how visual information from final dish images affects procedural state tracking performance. This analysis contributes to understanding the relative contributions of textual and visual modalities in procedural reasoning tasks.

2.4 World State Modeling in Natural Language Processing

Recent advances in natural language processing have increasingly emphasized the importance of world state modeling for comprehensive text understanding. Bosselut et al. [**?**] demonstrated that commonsense reasoning requires maintaining implicit world models, while Dalvi et al. [**?**] showed that procedural text understanding benefits from explicit state representations. These works highlight the critical gap between surface-level language understanding and deep procedural reasoning that requires tracking entity states through temporal sequences.

Building on these foundations, several recent efforts have explored explicit state tracking in procedural domains. The ProPara dataset [**?**] introduced systematic evaluation of entity state changes in procedural text, focusing on location and existence tracking across scientific processes. However, ProPara's synthetic nature and limited procedural complexity constrain its applicability to real-world cooking scenarios where ingredient transformations involve nuanced chemical and physical changes that are rarely explicitly described in natural language instructions.

2.5 State Tracking and World Model Learning

State tracking has been extensively studied across multiple domains, from dialogue systems to robotics and planning. In dialogue systems, state tracking involves maintaining conversation context and user goals [**?**], while in reinforcement learning, world models capture environment dynamics for planning [**?**]. However, these approaches typically operate with explicit state representations rather than the implicit tracking required in natural language procedural text.

The field of natural language processing has seen growing interest in commonsense reasoning and world knowledge, with benchmarks like CommonsenseQA [**?**] and ATOMIC [**?**] evaluating models' understanding of everyday events and their consequences. Yet these tasks focus on general knowledge rather than step-by-step procedural state evolution.

Recent work in instruction following has begun to address procedural understanding. ROTH et al. [**?**] introduced tasks requiring models to track entity states across procedural text, while ALFWorld [**?**] created embodied environments for language-based planning. However, these works operate in synthetic or simplified domains rather than the complex, implicit state transitions found in natural cooking recipes.

Our contribution lies in bridging the gap between explicit symbolic state tracking and implicit natural language understanding, providing a principled evaluation framework for assessing models' ability to maintain consistent world models throughout procedural execution in authentic, professionally curated text.

3 NHK Recipe Dataset

3.1 Data Source Selection

Our dataset is built upon recipes from "Minna no Kyo no Ryori" (Everyone's Today's Cooking), a prestigious educational cooking program by NHK (Japan Broadcasting Corporation). This source provides several key advantages for our state probing task: (1) Professional editorial oversight ensuring high-quality, consistent recipe formatting; (2) Standardized ingredient names and quantities that reduce annotation ambiguity; (3) HTML-encoded step references that create natural opportunities for reference resolution tasks; and (4) Professional photography of final dishes enabling multimodal evaluation.

Figure 2: Screenshot of the NHK Educational cooking website "Minna no Kyo no Ryori" showing the professional format and quality of the source recipes.

The initial dataset contains approximately 25,000 recipes scraped from the NHK website. These recipes exhibit remarkable consistency in structure, with clearly defined ingredient lists, step-by-step instructions, and embedded references to previous steps using standardized HTML formatting.

3.2 Annotation Protocol

To create explicit state annotations, we developed a comprehensive annotation protocol that captures ingredient transformations throughout cooking procedures. Our approach defines a step-wise world state $\mathcal{W}_t$ that evolves according to the update rule:

$$\mathcal{W}_{t+1} \leftarrow (\mathcal{W}_t \setminus \mathcal{U}) \cup \mathcal{V} \tag{1}$$

where $\mathcal{U}$ represents the set of items modified or consumed at step t, and $\mathcal{V}$ represents the resulting items produced by the transformation.

The annotation process follows these steps:

1. Initial State Definition: We define the initial world state $\mathcal{W}_0$ as the complete ingredient list $\mathcal{G}$ specified in the recipe.

2. Step-wise State Annotation: For each cooking step s_t, annotators examine the recipe title, ingredient list, and instruction sequence to determine the post-step world state $\mathcal{W}_t$. This includes identifying which ingredients are consumed, transformed, or combined.

3. Transformation Link Recording: We capture fine-grained transformation pairs (u, v) that map input items to output items at each step. This supports complex operations including one-to-many splits (e.g., "separate egg whites and yolks"), many-to-one combinations (e.g., "mix flour and sugar"), and temporary intermediate products.

4. Consistency Validation: We implement automated consistency checks to prevent unintended item creation or disappearance, ensuring that all transformations are properly tracked through input-output chains.

3.3 Sampling and Quality Control

From the full 25,000-recipe collection, we strategically sample 100 recipes to flatten the biased distribution of step-reference distances, creating approximately uniform coverage across different reference patterns. This sampling strategy ensures robust evaluation across various types of procedural dependencies.

Two human annotators performed the annotations using a spreadsheet-based interface with built-in consistency checks. The annotation process captures not only the final world states but also the detailed transformation graph that enables fine-grained analysis of ingredient flow through cooking procedures.

The annotated subset consists of 60 training recipes and 40 evaluation recipes, released publicly at https://huggingface.co/datasets/mashi6n/nhkrecipe-100-anno-1 to support future research in procedural understanding.

4 Task Definition

Our benchmark comprises three complementary tasks that systematically evaluate different aspects of procedural state tracking. Each task takes as input the recipe title, ingredient list, and step

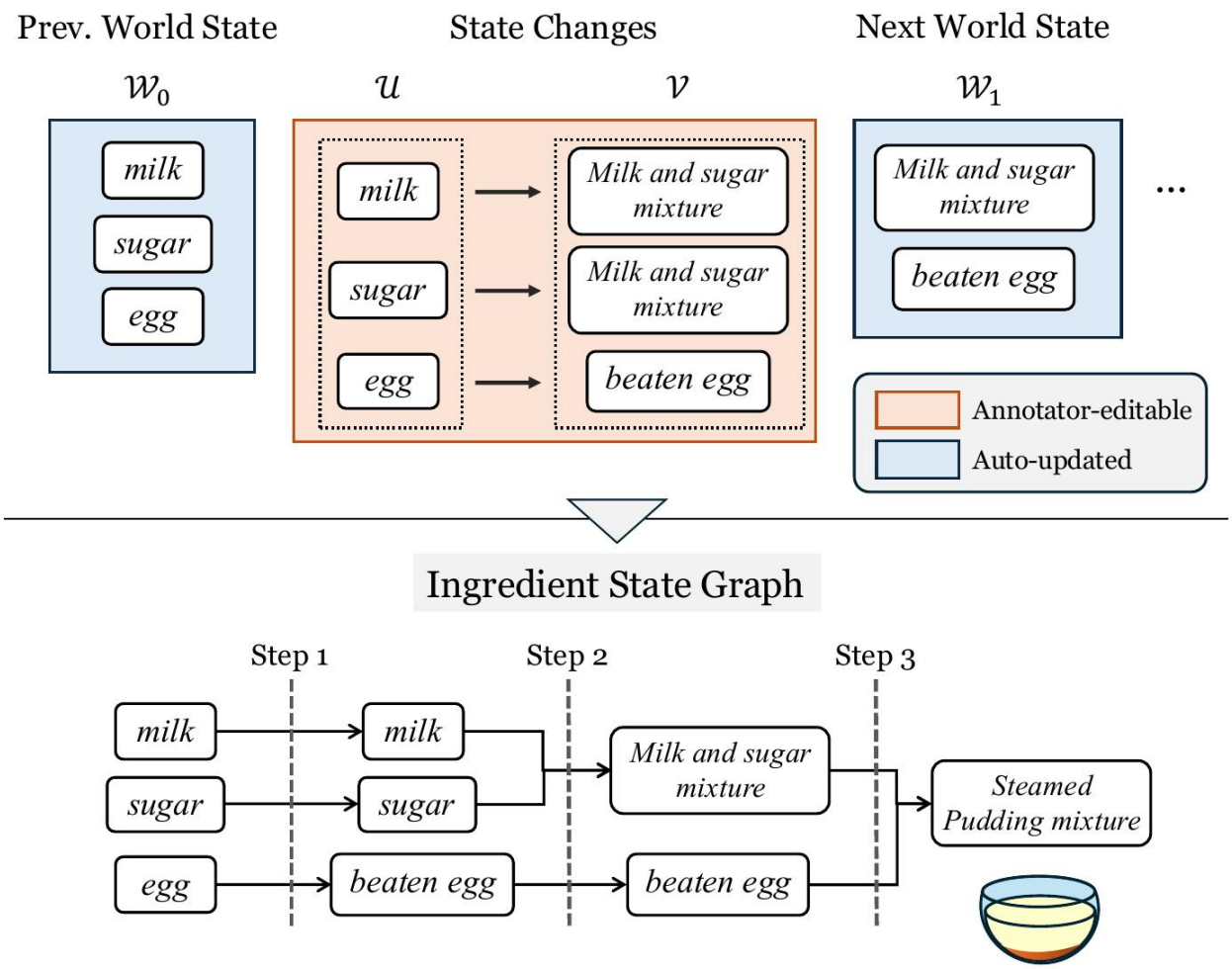

Figure 3: Annotation process (top) showing how annotators define world states for each step, and the resulting Ingredient State Graph (bottom) that captures fine-grained transformations including split/merge operations and intermediate products.

sequence, with multimodal variants additionally receiving the final dish image.

4.1 Task 1: Step Reference Resolution

Step Reference Resolution evaluates models' ability to understand implicit temporal dependencies in cooking instructions. Given a recipe where one step reference is masked, models must predict the correct step index being referenced.

Format: The task masks step references like "the chicken from step X" and asks models to identify the correct value of X.

Dataset Size: 33,106 instances generated from step references across the recipe collection.

Evaluation: Direct answer prediction without chain-of-thought or in-context learning. The chance rate is $\frac{1}{\tau-1}$ where τ is the current step index, averaging 0.348 across all instances.

This task assesses whether models can maintain consistent tracking of ingredient states and their associated preparation steps throughout recipe execution.

4.2 Task 2: Ingredient Usage Prediction

Ingredient Usage Prediction tests models' understanding of ingredient consumption patterns by asking whether a specific ingredient remains unused at a given step.

Format: Given a recipe and step index τ, models predict whether ingredient i is still in its original unused state.

Dataset Size: 346 instances covering diverse ingredient usage patterns.

Evaluation: Binary classification with 0.5 chance rate. This task directly evaluates models' ability to track which ingredients have been consumed or transformed.

4.3 Task 3: Ingredient Tracing

Ingredient Tracing represents the most challenging task, requiring models to identify which intermediate items at step τ contain a target ingredient. This evaluates understanding of complex ingredient

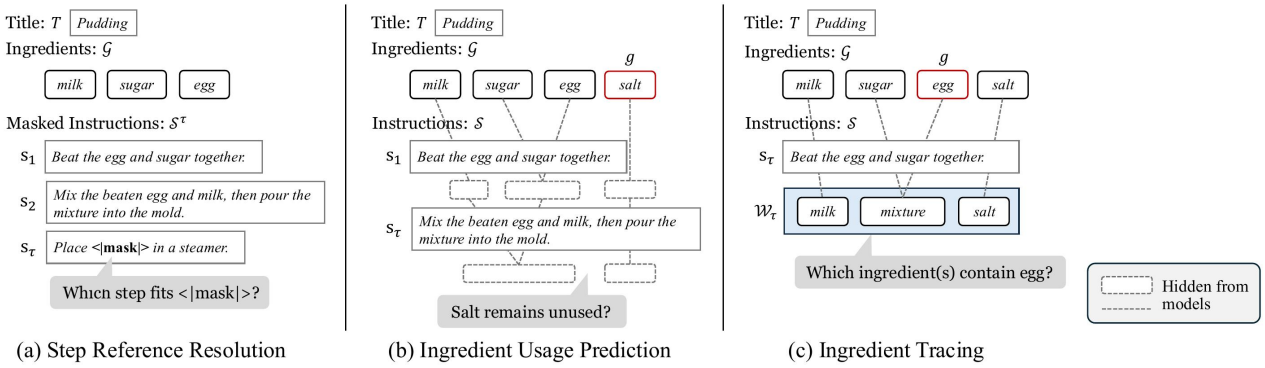

Figure 4: Overview of the three novel tasks we propose: Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing. Each task evaluates different aspects of procedural state understanding.

transformations including combinations, splits, and intermediate product creation.

Format: Given a target ingredient and step index, models select all items in the current world state $\mathcal{W}_\tau$ that contain the target ingredient.

Dataset Size: 333 instances featuring complex transformation scenarios.

Evaluation: Multi-choice selection with chance rate $\leq \frac{1}{|\mathcal{W}_\tau|}$, averaging approximately 0.193.

This task captures the most sophisticated aspects of state tracking, requiring models to understand how ingredients flow through multiple transformations while maintaining their essential identity.

4.4 Multimodal Extension

For large multimodal models (LMMs), we extend each task by prepending the final dish image as a visual token: "Dish image: <Image>". This enables evaluation of how visual information about the cooking outcome affects procedural state tracking performance.

The visual modality provides additional context about the expected final state, potentially helping models better understand the transformation trajectory throughout the cooking process.

5 Experiments and Results

5.1 Experimental Setup

We evaluate a comprehensive range of both language-only and multimodal models across different scales, from 7B to 72B parameters. Our model selection strategy targets diversity across architectural families and training methodologies. We include popular open-source models from the Llama [2, 4], Qwen [7], and Swallow series, as well as the closed-source GPT-4o model. The Swallow models are particularly relevant as they incorporate Japanese-specific training data, potentially offering advantages for our NHK recipe dataset.

For fine-tuning, we employ lightweight LoRA (Low-Rank Adaptation) training on the 60 annotated training recipes. This approach enables efficient adaptation while preserving the models' general capabilities. All evaluations use direct answer prediction without chain-of-thought reasoning or in-context learning examples, ensuring consistent evaluation conditions across all models and tasks.

5.1.1 Model Configuration and Training Details. Our LoRA implementation targets attention projection layers with rank $r = 16$ and scaling factor $\alpha = 32$, following established best practices for efficient parameter adaptation. We train for 3 epochs with a learning rate of 2×10^{-4} using AdamW optimizer with linear warmup over 10% of total steps, followed by cosine decay. Gradient clipping is applied at norm 1.0 to ensure stable training across different model scales.

For multimodal models, we freeze visual encoders and apply LoRA only to text-processing components, maintaining visual-textual alignment while enabling task-specific adaptation. Training examples are formatted as instruction-following prompts with clear task descriptions and direct answer targets. We employ early stopping based on validation performance to prevent overfitting on the limited training set.

To ensure robust evaluation, we implement stratified sampling across task types during training, maintaining balanced representation of reference resolution, usage prediction, and tracing examples. Our training data augmentation includes systematic permutation of ingredient ordering and minor textual variations to improve generalization. We also implement temperature scaling ($T = 0.7$) during inference to calibrate model confidence, particularly important for the multi-choice ingredient tracing task where overconfident predictions can significantly impact performance.

5.2 Language Model Results

Table 1 presents comprehensive results for language-only models across all three tasks. Several key patterns emerge from these results.

Scale Matters for Fine-tuning Effectiveness: Smaller models (< 10B parameters) often show performance degradation after fine-tuning. For example, Llama-3.1-8B drops from 0.324 to 0.269 average accuracy, with particularly sharp decline in Ingredient Tracing (0.462 → 0.111). In contrast, larger models consistently benefit from fine-tuning, with Llama-3.1-70B improving from 0.558 to 0.720 average accuracy.

Step Reference Resolution Shows Largest Gains: Fine-tuning provides dramatic improvements in Step Reference Resolution across most models. Llama-3.1-70B improves from 0.224 to 0.708,

Table 1: Experimental results for LLMs. "Step Ref.", "Ingr. Usage", and "Ingr. Trace" denote Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing, respectively.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Llama-3.1-8B	0.002	0.509	**0.462**	**0.324**
+ Fine Tuning	**0.184**†	**0.512**	0.111	0.269
Llama-3.1-70B	0.224	0.616	**0.835**	0.558
+ Fine Tuning	**0.708**†	**0.705**†	0.748	**0.720**
Llama-3.3-70B	0.595	0.520	0.844	0.653
+ Fine Tuning	**0.629**†	**0.566**†	**0.856**	**0.684**
Swallow-3.1-8B	0.330	**0.538**	**0.553**	**0.474**
+ Fine Tuning	**0.349**†	0.526	0.438	0.438
Swallow-3.1-70B	0.761	0.757	0.793	0.770
+ Fine Tuning	**0.809**†	**0.884**†	**0.817**	**0.837**
Swallow-3.3-70B	0.433	0.702	**0.880**	0.672
+ Fine Tuning	**0.495**†	**0.827**†	0.862	**0.728**
Qwen-2.5-7B	**0.199**	0.509	0.670	**0.459**
+ Fine Tuning	0.188	0.509	0.670	0.456
Qwen-2.5-14B	0.389	0.610	**0.763**	0.587
+ Fine Tuning	**0.568**†	**0.662**†	0.754	**0.661**
Qwen-2.5-32B	0.507	0.757	**0.817**	0.694
+ Fine Tuning	**0.659**†	**0.871**†	0.793	**0.774**
Qwen-2.5-72B	0.840	0.725	0.700	0.755
+ Fine Tuning	**0.871**†	**0.861**†	**0.712**	**0.815**
GPT-4o	0.776	0.812	0.883	0.824

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

and Swallow-3.1-70B from 0.761 to 0.809. This suggests that explicit state annotation helps models learn temporal reference patterns.

Model Architecture Influences Performance: The Qwen series shows consistently strong performance, with Qwen-2.5-32B achieving 0.774 average accuracy after fine-tuning. Japanese-trained Swallow models also perform well, with Swallow-3.1-70B reaching 0.837 average accuracy—competitive with the closed-source GPT-4o at 0.824.

5.3 Multimodal Model Results

Table 2 shows results for large multimodal models (LMMs) that incorporate final dish images alongside recipe text.

Visual Information Provides Substantial Benefits: Multimodal models demonstrate significant improvements over their language-only counterparts. Qwen-2.5-VL-72B achieves 0.919 average accuracy after fine-tuning, substantially outperforming even the best language-only models.

Ingredient Usage Prediction Benefits Most from Visual Input: The multimodal Qwen-2.5-VL-72B achieves 0.945 accuracy on Ingredient Usage Prediction after fine-tuning, suggesting that visual information about the final dish helps models understand ingredient consumption patterns.

Table 2: Experimental results for LMMs. All tasks and model training are conducted with images.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Qwen-2.5-VL-7B	0.094	0.497	**0.523**	**0.371**
+ Fine Tuning	**0.151**†	**0.500**	0.234	0.295
Qwen-2.5-VL-32B	0.239	0.740	**0.745**	0.575
+ Fine Tuning	**0.280**†	**0.812**†	0.655	**0.582**
Qwen-2.5-VL-72B	0.883	0.662	0.886	0.810
+ Fine Tuning	**0.919**†	**0.945**†	**0.892**	**0.919**

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

Consistent Fine-tuning Benefits: Unlike language-only models, all multimodal models show improvements from fine-tuning, indicating that the combination of visual and textual information creates more robust learning targets.

5.4 Error Analysis

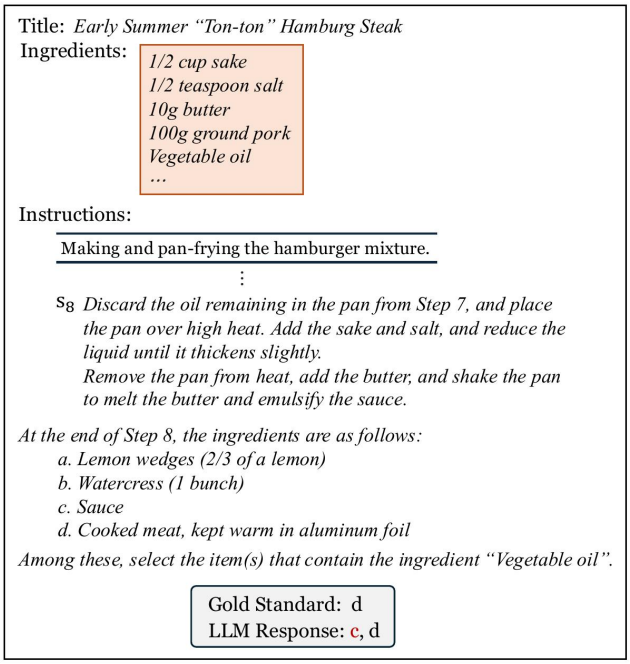

Figure 5: Failure Case 1: Model struggles with ingredient subtraction and parallel operations where multiple ingredients are processed simultaneously.

Analysis of model failures reveals persistent challenges in several areas:

Ingredient Subtraction: Models struggle when ingredients need to be partially removed or consumed, often failing to track

Title: *Early Bonito "Arare-yaki"*

Ingredients:

100 ml soy sauce (for marinade)
50 ml sake (for marinade)
...
250 g bonito
Shredded spring cabbage

Instructions:

s_1 *Mix the ingredients for the marinade, and set aside a small amount to use as a dipping sauce. Place the remaining marinade in a plastic bag and add the bonito.*

⋮

s_4 *Remove the bonito from the bag and pat it dry.Coat the entire surface evenly with crushed rice crackers (arare).*

s_5 *Heat 2 to 3 tablespoons of rice oil in a frying pan over high heat.Add the coated bonito (Step 4) and quickly sear both sides until crispy.*

At the end of Step 5, the ingredients are as follows:

a. Remaining marinade
b. Fried bonito
c. Shredded spring cabbage
...

Among these, select the item(s) that contain the ingredient "soy sauce".

Gold Standard: a, b
LLM Response: a

Figure 6: Failure Case 2: Models have difficulty tracking complex transformations involving splitting operations and temporary intermediate products.

remaining quantities correctly. For instance, when a recipe calls for "half the flour" in step 3 and "remaining flour" in step 7, models frequently lose track of the quantity partition.

Parallel Operations: When multiple ingredients undergo simultaneous transformations, models frequently lose track of individual state transitions. This is particularly problematic in steps like "while the chicken cooks, prepare the sauce," where temporal overlap creates complex dependencies.

Complex Transformations: Splitting operations (e.g., separating eggs) and temporary intermediate products challenge even the best-performing models. The creation of ephemeral states that exist only briefly before further transformation appears to exceed current attention mechanisms' capacity for state tracking.

These failure modes highlight fundamental limitations in current models' procedural reasoning capabilities and suggest directions for future improvement in compositional state representation.

5.5 Computational Requirements and Reproducibility

Our experimental setup requires significant computational resources to ensure comprehensive evaluation across model families. Training is conducted on NVIDIA A100 80GB GPUs, with larger models (70B+ parameters) requiring 4-8 GPUs using data parallel training. Memory optimization through gradient checkpointing and mixed-precision training (FP16) enables efficient utilization of available hardware resources.

Fine-tuning duration varies substantially by model size: smaller models (7B-13B) complete training in 2-4 hours, while larger models (70B+) require 12-24 hours for convergence. We implement automated checkpointing every 500 steps to ensure training resilience across extended training sessions.

For inference evaluation, we batch process examples with adaptive batch sizing based on model memory requirements. Temperature sampling ($T = 0.7$) and nucleus sampling ($p = 0.9$) provide consistent generation across different model architectures. We conduct three independent runs with different random seeds for each configuration, reporting mean performance with standard deviations to ensure statistical robustness.

All experimental configurations, hyperparameters, and evaluation scripts are made publicly available alongside the dataset to support reproducible research. Our evaluation framework supports both local deployment and distributed inference across multiple GPUs, with automatic model parallel strategies for larger architectures.

5.6 Statistical Significance

We assess statistical significance using one-sided McNemar tests comparing fine-tuned models against their zero-shot baselines. Results marked with † in Tables 1 and 2 indicate statistically significant improvements at $p < 0.01$, demonstrating that fine-tuning provides genuine benefits beyond random variation.

5.7 Discussion and Limitations

Our findings reveal important limitations in current procedural understanding. The scale dependency observed suggests fundamental architectural constraints in smaller models' capacity for maintaining complex state representations. Additionally, the persistent challenges with parallel operations highlight the need for explicit memory mechanisms in future model designs. While our dataset provides valuable insights, the annotation scale remains limited, and expansion to other procedural domains would strengthen generalization claims.

6 Conclusion

We have introduced NHK Recipe State Probing, the first benchmark to systematically evaluate models' ability to track implicit state changes in procedural text. Our work makes several important contributions to the field of grounded procedural understanding.

Novel Dataset and Annotation Framework: We present a high-quality dataset of 100 professionally curated recipes with explicit per-step world-state annotations and fine-grained ingredient transformation links. This dataset provides the first systematic resource for evaluating procedural state tracking in cooking domains.

Targeted Evaluation Tasks: Our three complementary tasks—Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing—directly assess different aspects of state tracking that are fundamental to procedural understanding but underexplored in existing benchmarks.

Scale-dependent Learning Patterns: Our comprehensive evaluation reveals that fine-tuning effectiveness depends critically on model scale. While smaller models (< 10B parameters) often degrade with fine-tuning, larger models show substantial and consistent

improvements. This finding has important implications for future research in procedural understanding.

Multimodal Advantages: Large multimodal models demonstrate superior performance, with visual information from final dish images providing complementary signals that enhance procedural state tracking. This suggests promising directions for multimodal approaches to procedural reasoning.

Persistent Challenges: Our error analysis reveals fundamental limitations in current models' ability to handle complex transformations, parallel operations, and ingredient subtraction. These challenges point to important areas for future development in procedural reasoning capabilities.

The public release of our dataset and evaluation framework at https://huggingface.co/datasets/mashi6n/nhkrecipe-100-anno-1 aims to foster continued research in grounded procedural understanding. We believe that explicit state modeling, as demonstrated in our work, represents a crucial step toward developing language models with deeper understanding of real-world procedural knowledge.

Future work should explore several promising directions: scaling annotation efforts to larger recipe collections through semi-automated labeling techniques, extending state probing to other procedural domains such as scientific protocols and assembly instructions, and developing specialized model architectures with explicit memory modules for state tracking. Additionally, investigating cross-lingual transfer learning could leverage our Japanese recipe annotations for procedural understanding in other languages. Our benchmark provides a foundation for these investigations and a reliable evaluation framework for measuring progress in procedural understanding research.

7 Acknowledgements

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References

[1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. 2025. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923* (2025).

[2] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, and Austen Gregerson et al. 2024. The Llama 3 Herd of Models. arXiv:2407.21783 [cs.AI] https://arxiv.org/abs/2407.21783

[3] Amaia Salvador, Nicholas Hynes, Yusuf Aytar, Javier Marin, Ferda Ofli, Ingmar Weber, and Antonio Torralba. 2017. Learning Cross-Modal Embeddings for Cooking Recipes and Food Images. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 3068–3076. doi:10.1109/CVPR.2017.327

[4] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288* (2023).

[5] Te-Lin Wu, Alex Spangher, Pegah Alipoormolabashi, Marjorie Freedman, Ralph Weischedel, and Nanyun Peng. 2022. Understanding Multimodal Procedural Knowledge by Sequencing Multimodal Instructional Manuals. In *ACL (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 4525–4542. doi:10.18653/v1/2022.acl-long.310

[6] Semih Yagcioglu, Aykut Erdem, Erkut Erdem, and Nazli Ikizler-Cinbis. 2018. RecipeQA: A Challenge Dataset for Multimodal Comprehension of Cooking Recipes. In *EMNLP*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 1358–1368. doi:10.18653/v1/D18-1166

[7] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. 2024. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115* (2024).